SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	**Slot A**	**Date:**	**14 - 08 - 2026**

Code of Conduct

I ***Mohammed Rayan Shaikh - 192424059*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___________________________

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:

Incorrect interpretation of ambiguous sentences
Failure to handle subject–verb agreement
Inefficient parsing for long queries
Example user query:
"Show me the transactions with the card from last month"
a) Analyze the ambiguity present in the given sentence using CFG
b) Evaluate the limitations of CFG in handling real-world conversational queries
c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing
d) Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:

Backtracking issues
Poor handling of incomplete or ambiguous input
Difficulty in real-time response
Example command:
"Book a flight to Delhi with a window seat"
a) Analyze how the given sentence can lead to multiple parse structures
b) Analyze the limitations of top-down parsing in this real-time application
c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs
d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.

The system must handle:
Complex and lengthy medical sentences
Ambiguity in sentence structure
Accurate subject-verb agreement
Relationships between medical actions and entities (sub-categorization)
Real-time processing for hospital use
Example input:
"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."
The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.
a) Design a complete NLP architecture integrating:
CFG for syntactic structure
Probabilistic CFG (PCFG) for ambiguity resolution
Feature Structures for agreement handling
Efficient parsing techniques
b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).
c) Create a method to:
Incorporate sub-categorization frames for medical verbs

Handle real-time data processing
Improve accuracy and scalability in a hospital environment

Rubrics for Calculation

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. AI-Powered Customer Support Chatbot for Banking Services

a) Analyze the ambiguity present in the given sentence using CFG

The given sentence is:

"Show me the transactions with the card from last month."

The sentence contains structural ambiguity, mainly because of the phrases "with the card" and "from last month".

The phrase "with the card" can be interpreted as describing the transactions, meaning the transactions were made using the card. Another possible interpretation is that "with the card" is an additional instruction associated with the request.

The phrase "from last month" can also be attached to the transactions or to the overall request.

The intended interpretation is:

ACTION = SHOW
OBJECT = TRANSACTIONS
PAYMENT_METHOD = CARD
TIME_PERIOD = LAST_MONTH

Therefore, the semantic representation is:

SHOW(Transactions, Card, Last_Month)

A simplified CFG can be represented as:

S → VP
VP → V NP
VP → V NP PP
NP → Det N
NP → NP PP
PP → P NP

CFG can generate multiple parse trees for the same sentence. Therefore, it may not always determine the intended interpretation.

b) Evaluate the limitations of CFG in handling real-world conversational queries

A basic Context-Free Grammar has several limitations when used in real-world conversational systems.

1. CFG can generate multiple interpretations for an ambiguous sentence.
2. Basic CFG does not assign probabilities to different interpretations.
3. CFG does not naturally understand conversational context.
4. Basic CFG does not effectively handle subject-verb agreement.
5. Long sentences can produce a large number of possible parse trees.

6. Incomplete user queries can be difficult to parse.
7. Speech recognition errors can cause parsing failures.
8. Banking applications require large domain-specific vocabularies.
9. Users may express the same intention using different sentence structures.
10. CFG alone does not understand the semantic meaning of words.

Therefore, a basic CFG is not sufficient for a large-scale banking chatbot.

c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing

A better solution is to combine PCFG, Feature Structures, and Earley Parsing.

A Probabilistic Context-Free Grammar assigns probabilities to grammar rules. When multiple parse trees are possible, the system can select the parse with the highest probability.

For example:

VP → V NP PP 0.60
VP → VP PP 0.40

The system can use these probabilities to select the most likely interpretation.

Feature Structures can represent grammatical information such as:

Person = Third
Number = Singular
Tense = Present

They can be used to ensure subject-verb agreement.

Earley Parsing is useful for ambiguous and incomplete sentences because it can maintain multiple possible parsing hypotheses efficiently.

The improved architecture can be:

User Query
↓
Preprocessing
↓
Tokenization
↓
Feature Extraction
↓
Earley Parser
↓
PCFG Probability Ranking
↓
Feature Agreement Checking
↓

Semantic Interpretation
↓
Banking Intent
↓
Response/Action

For the given query, the final semantic representation would be:

SHOW_TRANSACTIONS
Payment Method = Card
Time Period = Last Month

d) Justify how the proposed method improves accuracy and efficiency in an industry setting

The proposed approach improves the banking chatbot in several ways.

PCFG helps select the most probable interpretation when the sentence is ambiguous.

Feature Structures improve grammatical accuracy by checking properties such as number, person, and tense.

Earley Parsing reduces unnecessary repeated parsing and can handle ambiguous and incomplete input.

Domain-specific grammar can be created for banking terms such as account, balance, transaction, card, loan, transfer, and payment.

The structured interpretation can then be sent to the banking service.

For example:

“Show me the transactions with the card from last month.”

can be converted into:

Intent = SHOW_TRANSACTIONS
Payment Method = CARD
Time Period = LAST_MONTH

This improves intent recognition, response accuracy, and customer experience.

2. Voice Assistant Using Top-Down and Earley Parsing

a) Analyze how the given sentence can lead to multiple parse structures

The given command is:

“Book a flight to Delhi with a window seat.”

The phrase “with a window seat” creates an attachment ambiguity.

One interpretation is:

[Book [a flight to Delhi] [with a window seat]]

Here, “with a window seat” specifies the user's seat preference.

Another possible interpretation is:

[Book [a flight [to Delhi with a window seat]]]

Here, the phrase is treated as part of the description of the flight.

The intended semantic representation is:

INTENT = BOOK_FLIGHT
DESTINATION = DELHI
SEAT_PREFERENCE = WINDOW

A simplified parse structure is:

S
→ VP
→ V NP PP
→ Book + a flight to Delhi + with a window seat

Therefore, the parser must determine which interpretation is more appropriate.

b) Analyze the limitations of top-down parsing in this real-time application

Top-down parsing starts with the start symbol and attempts to generate the input sentence.

It has several limitations.

1. It may require backtracking when an incorrect grammar rule is selected.
2. Ambiguous sentences can result in multiple possible parsing paths.
3. Backtracking increases processing time.
4. It has difficulty handling incomplete commands.
5. Speech recognition may produce fragmented or incorrect input.

6. Real-time applications require fast response times.
7. Repeated parsing of similar structures can be inefficient.
8. Contextual interpretation is limited in basic top-down parsing.

For example, a user may say:

“Book a flight to Delhi…”

and pause before completing the command. A traditional top-down parser may not handle this partial input effectively.

c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs

Earley parsing is designed to handle general Context-Free Grammars and can maintain multiple parsing possibilities.

For the sentence:

“Book a flight to Delhi with a window seat.”

the parser can maintain different possible interpretations until sufficient information is available.

It can also process partial input such as:

“Book a flight to…”

and maintain the current parsing state while waiting for additional words.

Earley parsing provides:

1. Better ambiguity handling.
2. Support for incomplete input.
3. Reduced unnecessary backtracking.
4. Efficient chart-based parsing.
5. Support for general CFGs.
6. Ability to maintain multiple parsing hypotheses.

This makes Earley parsing more suitable for voice-based conversational applications.

d) Compare and analyze the performance of top-down vs Earley parsing

Feature	**Top-Down Parsing**	**Earley Parsing**
Ambiguity Handling	Limited	Good
Backtracking	Frequently required	Reduced
Partial Input	Poor	Better
Incomplete Commands	Limited	Good
General CFG Support	Limited depending on implementation	Yes

Multiple Parse Structures	Difficult	Supported
Real-Time Use	Suitable for simple grammar	Suitable for complex grammar
Memory Usage	Usually lower	Higher due to chart
Complex Sentences	Can become inefficient	More robust

For simple voice commands, top-down parsing can be fast and easy to implement.

However, for complex, ambiguous, or incomplete commands, Earley parsing is more appropriate.

A practical voice assistant architecture is:

Speech Input
↓
Speech Recognition
↓
Tokenization
↓
Earley Parser
↓
PCFG/Statistical Ranking
↓
Intent Detection
↓
Entity Extraction
↓
Action

For the given command, the final interpretation is:

BOOK_FLIGHT
Destination = Delhi
Seat Preference = Window

3. Healthcare NLP System Using CFG, PCFG and Feature Structures

Given sentence:

“The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.”

a) Design a complete NLP architecture

The healthcare NLP system should combine syntactic, probabilistic, morphological, and semantic processing.

The proposed architecture is:

Medical Report
↓
Text Preprocessing
↓
Sentence Segmentation
↓
Tokenization
↓
Medical Named Entity Recognition
↓
Feature Extraction
↓
CFG Parsing
↓
Earley Parsing
↓
PCFG Probability Ranking
↓
Feature Structure Checking
↓
Sub-Categorization Analysis
↓
Semantic Role Labeling
↓
Medical Knowledge Mapping
↓
Structured Medical Output

CFG is used to identify the basic syntactic structure.

Example grammar rules include:

S → NP VP
NP → Det N
NP → NP RelClause
VP → V NP

VP → V GerundPhrase
VP → VP Conjunction VP
RelClause → Pronoun VP
PP → P NP

PCFG is then used to assign probabilities to alternative parse structures. The most probable interpretation can be selected.

Feature Structures can store grammatical features such as:

Person = Third
Number = Singular
Tense = Present

For example:

Doctor → Number = Singular
Recommends → Number = Singular

Therefore, subject-verb agreement is satisfied.

Earley Parsing can be used for efficient parsing of long and ambiguous medical sentences.

b) Step-by-step workflow

Step 1: Input

The system receives:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

Step 2: Tokenization

The sentence is divided into:

The
doctor
who
reviewed
the
patient
last
week
recommends
starting
medication
and
scheduling
a

follow-up
visit
in
Chennai

Step 3: Medical Entity Recognition

The system identifies:

Doctor → Person/Medical Entity
Patient → Medical Entity
Medication → Treatment
Chennai → Location

Step 4: Syntactic Parsing

The sentence can be divided into:

Subject:

The doctor who reviewed the patient last week

Main Verb:

recommends

Main Actions:

starting medication
scheduling a follow-up visit in Chennai

The relative clause is:

who reviewed the patient last week

This clause modifies “doctor”.

Step 5: Feature Agreement

The subject is:

Doctor = Singular

The main verb is:

recommends = Third Person Singular Present

Therefore:

Doctor + recommends

has correct subject-verb agreement.

Step 6: Relative Clause Analysis

The system identifies:

Doctor → reviewed → Patient
Time = Last Week

Therefore:

REVIEW(Doctor, Patient, Last_Week)

Step 7: Main Action Analysis

The main action is:

RECOMMEND

The doctor recommends two actions:

1. Starting medication.
2. Scheduling a follow-up visit.

Therefore:

RECOMMEND(Doctor, Start(Medication))

RECOMMEND(Doctor, Schedule(Follow_Up_Visit))

Step 8: Location Extraction

The phrase:

“in Chennai”

provides the location of the follow-up visit.

Therefore:

LOCATION(Follow_Up_Visit, Chennai)

Step 9: Final Semantic Representation

The structured representation is:

REVIEW(Doctor, Patient, Last_Week)

RECOMMEND(Doctor, Start(Medication))

RECOMMEND(Doctor, Schedule(Follow_Up_Visit))

LOCATION(Follow_Up_Visit, Chennai)

Step 10: Final Structured Output

Information	**Extracted Value**
Agent	Doctor
Reviewed Entity	Patient
Review Time	Last Week
Main Action	Recommend
Treatment	Starting Medication
Follow-up Action	Schedule Follow-up Visit
Location	Chennai
Diagnosis	Not Specified

The system should not generate a diagnosis because the given sentence does not explicitly mention one.

c) Method for sub-categorization, real-time processing, accuracy and scalability

Sub-Categorization Frames

Medical verbs can be stored with their expected arguments.

For example:

RECOMMEND → Doctor + Treatment/Action

PRESCRIBE → Doctor + Medicine + Patient

REVIEW → Doctor + Patient/Report

MONITOR → Doctor/Nurse + Patient

SCHEDULE → Agent + Appointment

DIAGNOSE → Doctor + Patient + Condition

For example:

“The doctor prescribed medicine to the patient.”

can be represented as:

PRESCRIBE(Doctor, Medicine, Patient)

This prevents the system from assigning inappropriate arguments to medical verbs.

Real-Time Processing

For real-time hospital applications, the system can use:

Hospital Input
↓
Message Queue
↓
Preprocessing
↓
NER + POS Tagging
↓
Efficient Parsing
↓
Semantic Analysis
↓
Structured Medical Data
↓
Hospital Database

The system can improve response time by using:

1. Incremental parsing.
2. Efficient Earley parsing.
3. Caching frequently used grammar rules.
4. Parallel document processing.
5. Asynchronous processing.
6. Domain-specific medical dictionaries.
7. Lightweight processing for real-time alerts.
8. Batch processing for historical medical reports.

Accuracy Improvements

Accuracy can be improved by:

1. Using medical-domain CFG rules.
2. Using PCFG for ambiguity resolution.
3. Using Feature Structures for agreement.
4. Using medical Named Entity Recognition.
5. Using Semantic Role Labeling.
6. Using medical ontologies.
7. Using sub-categorization frames.
8. Handling negation.
9. Handling temporal information.
10. Using domain-specific language models.

Scalability

A scalable hospital NLP system can use separate processing components for:

Real-Time Reports
Historical Reports
Emergency Alerts
Clinical Documents

The system can process independent documents in parallel and store the structured semantic information in a centralized medical database.

Conclusion

Basic CFG alone is not sufficient for large-scale banking, voice assistant, and healthcare NLP systems. A combination of CFG, PCFG, Feature Structures, Earley Parsing, semantic analysis, and domain-specific knowledge provides better ambiguity resolution, grammatical accuracy, efficiency, and scalability.